

Assignment: Optimization

Linear Programming Problem up to the Duality

Course: Optimization

Programme: B.Tech

Submission

Date:

Name: _____

Roll No.: _____

Question 1

Q1. What is a Linear Programming Problem?

Q2. Write down the mathematical form of a general L.P.P.

Q3. What is the relation between the optimum value of a maximization problem and a minimization problem?

Q4. Write down the standard form of a general L.P.P.

Q5: A factory produces two products A and B . Each unit of product A requires 2 hours of moulding, 3 hours of grinding, and 4 hours of polishing. Each unit of product B requires 4 hours of moulding, 2 hours of grinding, and 2 hours of polishing.

The moulding machine is available for 20 hours, the grinding machine for 24 hours, and the polishing machine for 13 hours. The profit is ₹5 per unit of A and ₹3 per unit of B . Formulate the problem as an LPP to maximize the profit

Q6: An electronics workshop produces two types of circuit boards: a standard board and a premium board. A standard board requires 2 hours of machining and 1 hour of testing. A premium board requires 1 hour of machining and 3 hours of testing. Each week, at most 120 machining hours and 150 testing hours are available. The profit from a standard board is Rs. 50 and that from a premium board is Rs. 60. The workshop can produce at most 40 premium boards per week. A dietician decides a certain minimum intake of vitamins A , B and C for a family. The minimum daily requirements of vitamins A , B , and C are 30, 20, and 16 units, respectively.

For supplying these vitamins, two types of foods X and Y are available. One gram of food X provides 7, 5, and 2 units of vitamins A , B , and C , respectively. One gram of food Y provides 2, 4, and 8 units of vitamins A , B , and C , respectively. The cost of food X is ₹ 2 per gram and the cost of food Y is ₹ 1 per gram.

Determine how many grams of each food should be purchased per day so that the total food expense is minimum. Formulate the problem as a linear programming problem.

Q7: A firm can produce two products A and B during a given period of time. Each of these products requires four different operations, namely, Grinding, Turning, Assembling, and Testing.

The requirement in hours per unit of manufacture of these products is given in the following table:

Product	Grinding	Turning	Assembling	Testing
A	1	3	4	5
B	2	1	3	4

The available capacities of these operations during the given time period are 30 hours for grinding, 60 hours for turning, 200 hours for assembling, and 200 hours for testing.

The profit on each unit of product A is ₹ 3, while the profit on each unit of product B is ₹ 2.

Formulate the problem as a linear programming model to maximize the profit, assuming that the firm can sell all the items it produces at the prevailing market price.

Q8: Four different metals, namely iron, copper, zinc, and manganese, are required to produce three commodities A , B , and C . To produce one unit of commodity A , 40 kg of iron, 30 kg of copper, 7 kg of zinc, and 4 kg of manganese are required. To produce one unit of commodity B , 70 kg of iron, 14 kg of copper, and 9 kg of manganese are required. To produce one unit of commodity C , 50 kg of iron, 18 kg of copper, and 8 kg of zinc are required. The total available quantities of metals are: 1 metric ton of iron, 5 quintals of copper, and 2 quintals each of zinc and manganese. The profits are ₹ 300, ₹ 200, and ₹ 100 on selling one unit of A , B , and C , respectively. Formulate the problem mathematically as a linear programming problem.

Q9. Solve the following linear programming problems by the graphical method:

1. Maximize $Z = 5x_1 + 7x_2$
subject to

$$\begin{aligned}x_1 + x_2 &\leq 4, \\3x_1 + 8x_2 &\leq 24, \\10x_1 + 7x_2 &\leq 35, \\x_1, x_2 &\geq 0.\end{aligned}$$

2. Maximize $Z = 5x_1 + 7x_2$
subject to

$$\begin{aligned}x_1 + x_2 &\leq 4, \\3x_1 + 8x_2 &\leq 24, \\10x_1 + 7x_2 &\leq 35, \\x_1, x_2 &\geq 0.\end{aligned}$$

3. Maximize $Z = 8x_1 + 7x_2$
subject to

$$\begin{aligned}3x_1 + x_2 &\leq 66000, \\x_1 + x_2 &\leq 45000, \\x_1 &\leq 20000, \\x_2 &\leq 40000, \\x_1, x_2 &\geq 0.\end{aligned}$$

Q: Define

1. a feasible solution,

2. Optimum solution,
3. Basic feasible solution.
4. Optimal Basic feasible solution.
5. Non degenerate Basic Feasible
6. Degenerate Basic feasible solution.

Q:: Differentiate between a feasible solution and a basic feasible solution of an LPP.

Q:: Find all the basic solutions of the following system:

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 4, \\2x_1 + x_2 + 5x_3 &= 5,\end{aligned}$$

and prove that they are non-degenerate.

Q:: Find all the basic feasible solutions of the system

$$\begin{aligned}2x_1 + 6x_2 + 2x_3 + x_4 &= 3, \\6x_1 + 4x_2 + 4x_3 + 6x_4 &= 2,\end{aligned}$$

subject to $x_i \geq 0$, $i = 1, 2, 3, 4$, and determine the associated general convex combination of the extreme point solutions.

Q: Show that $x_1 = 2$, $x_2 = 3$, $x_3 = 1$ is a feasible solution of the following linear programming problem:

$$\text{Maximize} \quad Z = x_1 + 2x_2 + 4x_3$$

subject to

$$\begin{aligned}2x_1 + x_2 + 4x_3 &= 11, \\3x_1 + x_2 + 5x_3 &= 14,\end{aligned}$$

and $x_1, x_2, x_3 \geq 0$.

Q: Give three advantages of the simplex method over the graphical method.

Q: Define Slack variable and Surplus variable.

Q: When are artificial variables introduced in a Linear Programming Problem

Q: State the fundamental theorem of LPP.

Q: Let $x_1 = 2$, $x_2 = 4$, $x_3 = 1$ be a feasible solution of the system

$$\begin{aligned}2x_1 - x_2 + 2x_3 &= 2, \\x_1 + 4x_2 &= 18,\end{aligned}$$

with $x_1, x_2, x_3 \geq 0$. Reduce the given feasible solution to a basic feasible solution.

Q: Give the computational procedure for the simplex method in linear programming

Q: Differentiate between a primal problem and its dual problem in linear programming.

Q: Define primal and dual linear programming problems. Give an example of a primal problem and write its corresponding dual problem.

Q: Write the dual of the following problem:

$$\begin{array}{ll}\text{Minimize} & Z = 3x_1 + x_2, \\ \text{subject to} & 2x_1 + 3x_2 \geq 2, \\ & x_1 + x_2 \geq 1, \\ & x_1, x_2 \geq 0.\end{array}$$

Q: Write the dual of the following problem:

$$\begin{array}{ll}\text{Minimize} & Z = 2x_2 + 5x_3, \\ \text{subject to} & x_1 + x_2 \geq 2, \\ & 2x_1 + x_2 + 6x_3 \leq 6, \\ & x_1 - x_2 + 3x_3 = 4, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Q: Give the dual of the following linear programming problem:

$$\begin{array}{ll}\text{Maximize} & Z = 2x_1 + 3x_2 + x_3, \\ \text{subject to} & 4x_1 + 3x_2 + x_3 = 6, \\ & x_1 + 2x_2 + 5x_3 = 4, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Q: Find the dual of the following linear programming problem:

$$\begin{array}{ll}\text{Minimize} & Z = x_1 + x_2 + x_3, \\ \text{subject to} & x_1 - 3x_2 + 4x_3 = 5, \\ & x_1 - 2x_2 \leq 3, \\ & 2x_2 - x_3 \geq 4, \\ & x_1, x_2 \geq 0, \\ & x_3 \text{ is unrestricted in sign.}\end{array}$$